Malath

An integrated smart system for guidance and safe shelter during emergencies

Engineering Field: Safe Shelter

[Team name] · [Team members]

The Problem

During fires in large buildings, people struggle to find safe evacuation routes and to know where the danger is.

Smoke and poor visibility

Smoke blocks sight and makes it hard to reach the exits.

Unfamiliar building layout

Visitors do not know the exits or the alternative routes.

Most affected: elderly people, visitors and those unfamiliar with the place

Groups that need clearer guidance and a nearby shelter.

The Solution: Four Integrated Layers

Smart light guidance that leads people to safety, and shelter capsules for those who cannot evacuate.

A

Sensing

Smoke, heat and CO gas sensors locate the danger and track how it develops.

B

Control Unit

Closes the affected route and recalculates the shortest safe path automatically.

C

Guidance

Screens and LED signals: green for safe, red for danger, plus voice guidance.

D

Shelter Capsules

Temporary refuge for those unable to evacuate, linked to Civil Defense.

How It Works

▶ Intro video



رصد الحريق وتحديد موقعه
وإطلاق التنبيهات الصوتية والمرئية.



شاشات تحدد موقع الخطر،
وإشارات خضراء للمسارات الأمنة
وحمراء للاتجاهات المؤدية للخطر.



توجيه الأشخاص إلى كبسولات
الإيواء المؤقت وانتظار فرق الطوارئ.

4 Notify Civil Defense Locations of people inside the capsules are sent to rescue teams.

Applicability

Where

Hospitals, malls, universities, hotels, towers and airports.

How

Gradual installation in existing buildings without demolition, starting with the highest-risk areas.

Compatibility

Works with existing alarm systems and takes into account fire protection requirements in the Saudi Building Code.

Impact

Shorter evacuation time and less confusion, protection for the most vulnerable, and help for rescue teams.